INCH-POUND
MIL-DTL-24565A(SH)
8 December 2020
SUPERSEDING
MIL-H-24565(SH)
3 March 1978

DETAIL SPECIFICATION

HOSE AND FITTINGS, OIL AND TURBINE FUEL, SMOOTH BORE, CORRUGATED COVER, 7-INCH

Reinstated after 8 December 2020 and may be used for new and existing designs and acquisitions.

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a 7-inch, smooth bore hose with corrugated cover, including fittings, used for receiving oil and turbine fuel with the probe fueling system receivers.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-A-8625 - Anodic Coatings for Aluminum and Aluminum Alloys

(Copies of this document are available online at https://quicksearch.dla.mil.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ASQ/ANSI Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at https://webstore.ansi.org.)

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 4720

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ASTM INTERNATIONAL

ASTM B221 - Standard Specification for Aluminum and Aluminum-Alloy Extruded Bars, Rods, Wire, Profiles, and Tubes

ASTM B247 - Standard Specification for Aluminum and Aluminum-Alloy Die Forgings, Hand Forgings, and Rolled Ring Forgings

ASTM D380 - Standard Test Methods for Rubber Hose

ASTM D381 - Standard Test Method for Gum Content in Fuels by Jet Evaporation

ASTM D412 - Standard Test Methods for Vulcanized Rubber and Thermoplastic Elastomers – Tension

ASTM D471 - Standard Test Method for Rubber Property - Effect of Liquids

ASTM D573 - Standard Test Method for Rubber – Deterioration in an Air Oven

(Copies of these documents are available online at www.astm.org.)

THE ASSOCIATION FOR HOSE AND ACCESSORIES DISTRIBUTION (NAHAD)

NAHAD Handbook - Design and Specification of Hose Assemblies

(Copies of this document are available at https://www.nahad.org/.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Construction. Hose dimensions shall conform dimensionally to figure 1. The hose shall have a rubber tube, reinforcing of nylon or polyester (applied as fabric, braid, or spiral wrap), a helix of round wire, additional nylon or polyester reinforcing, and a rubber cover.

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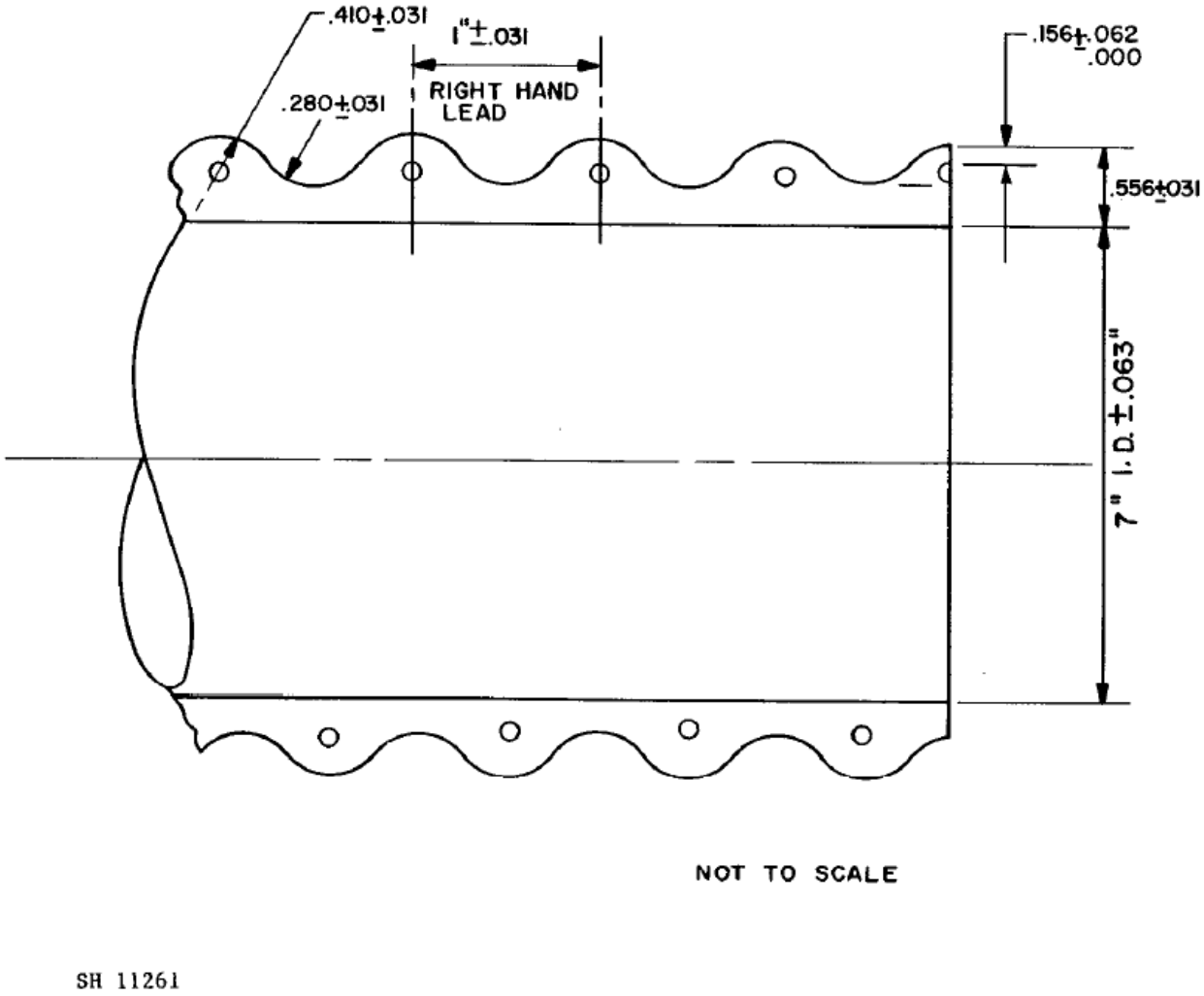


FIGURE 1. Convolution details of hose carcass.

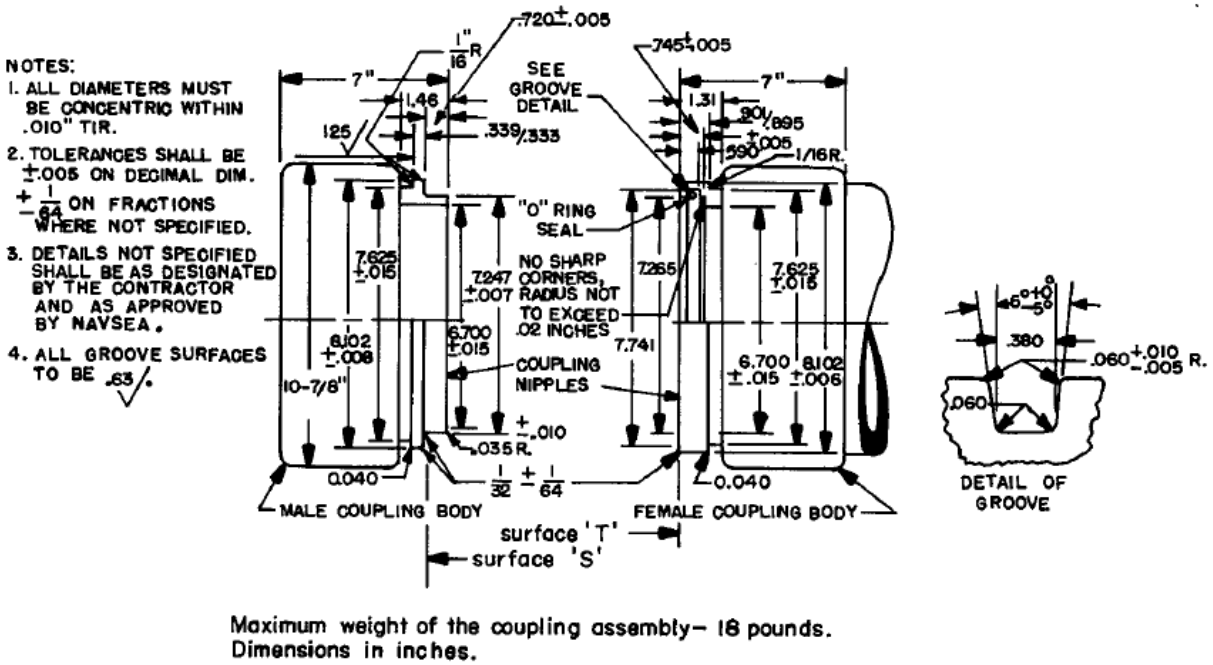
3.2 Wire. The helix wire shall be high strength metal, treated if necessary to prevent corrosion, having an outside diameter (o.d.) of 0.207 inch. The helix shall be applied in the direction of a right-hand thread having a pitch of 1 inch and shall be continuous to both ends of the hose.

3.3 Length.

3.3.1 Bulk hose. Bulk hose lengths shall be provided in 9-foot ±½-inch lengths. Hose may be fabricated in longer lengths for purposes of obtaining samples for testing (see 4.2.2).

3.3.2 Inspection length of hose assembly. An inspection length of hose assembly shall be 8 feet, 11⅜ inches ±½ inch between surfaces "S" and "T" (see figure 2).

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Maximum weight of the coupling assembly– 18 pounds.
Dimensions in inches.

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FIGURE 2. Reattachable coupling.

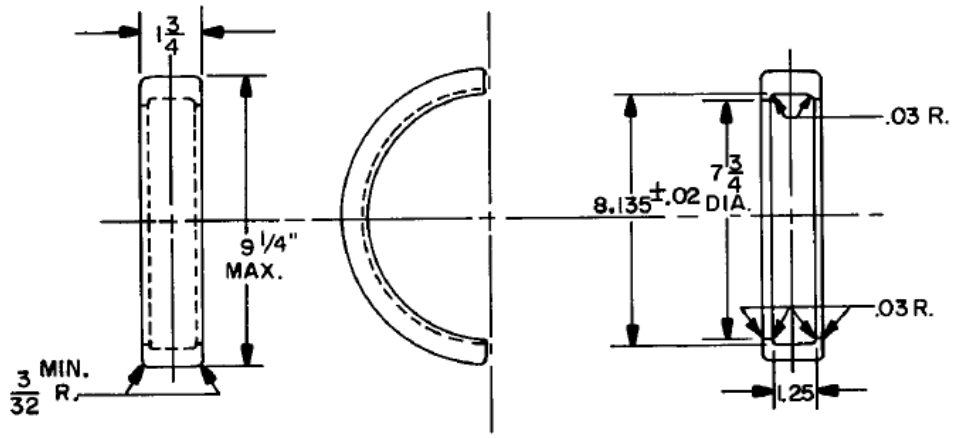
3.4. Tube and cover thickness. Thickness of the tube and cover shall be a minimum of 0.125 inch after buffing.

3.5 Diameter. The hose shall have a 7-inch ±0.063-inch nominal inside diameter (i.d.).

3.6. Weight. The maximum weight of the hose shall be 11.0 pounds per foot.

3.7 End fittings. End fittings shall be fabricated of aluminum in accordance with ASTM B221 or ASTM B247 and anodized in accordance with type I, MIL-A-8625. End fittings shall be reusable. The socket shall be of the segmented type, and the segments shall clamp over the spiral wire reinforcement helix in the hose carcass without special construction on the hose ends. Hose fitting and split clamp are shown on figure 2 and figure 3.

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FIGURE 3. Split clamp.

3.8 Flexibility. When subjected to the flexibility test as specified in 4.5.10, bending and torqueing the hose shall not reduce the minor diameter of the resulting ellipse to less than 5 inches. The wire helix shall not break or break through the tube or cover of the hose.

3.9 Burst pressure. When tested as specified in 4.5.7, the hose shall not leak or burst, nor shall the end fittings come off the hose at a pressure up to 550 pounds per square inch (lb/in^2).

3.10 Proof pressure. When tested as specified in 4.5.5 with a proof pressure of 250 lb/in^2, there shall be no leakage or damage to the hose.

3.10.1 Dimensional stability. The hose shall not increase in length more than 8 percent, nor shall the hose twist more than 1 degree per foot of hose when subjected to the proof test of 4.5.5.

3.11 Rubber components. The properties of the rubber in the tube, cover, and friction of the hose shall be in accordance with table I. The rubber used in the tube and friction shall be a copolymer of butadiene and acrylic nitrite. The rubber used in the cover shall be a blend of a copolymer of butadiene and acrylic nitrite and polyvinylchloride.

TABLE I. Requirements for rubber components in the hose.

	Tube	Cover	Friction	Test paragraph
Initial properties:				
Tensile strength, lb/in^2, minimum	2,600	1,600	--	4.5.9.1
Ultimate elongation	300	350	--	4.5.9.1
Adhesion, pounds per inch, minimum	15	15	15	4.5.9.5
	Tube	**Cover**	**Friction**	**Test paragraph**
Properties after immersion in test fluid:				
Tensile strength, lb/in^2, minimum	1,400	800	--	4.5.9.2

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TABLE I. Requirements for rubber components in the hose – Continued.

Ultimate elongation, percent minimum	200	200	--	4.5.9.2
Volume increase, percent maximum (no shrinkage permitted)	30	30	30	4.5.9.4
Adhesion, pounds per inch, minimum	10	10	10	4.5.9.6
Properties after oven aging:				
Tensile strength, lb/in^2, minimum	2,000	1,450	--	4.5.9.3
Ultimate elongation, percent minimum	250	250	--	4.5.9.3
Nonvolatile extract from tube, percent maximum	3.0	--	--	4.5.9.7
Phosphate plasticizer in tube, qualitative determination	Absent	--	--	4.5.9.8
Resistance of cover to ozone	--	No cracking	--	4.5.9.9 through 4.5.9.9.3

3.12 Hose and end fitting marking.

3.12.1 Hose. The hose shall have two rubber labels vulcanized into the cover, with one label located 2 feet from each end. Labels shall contain the following:

a. Manufacturer's name or trademark.

b. Month and year of manufacture.

c. Specification number.

d. "7-inch receiver hose".

3.12.2 End fittings. End fittings shall be stamped with the following:

a. Manufacturer's name or trademark.

b. Specification number.

c. Manufacturer's part number.

d. Nipples and segments shall contain the manufacturer's piece number.

3.13 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

4. VERIFICATION

4.1 Conformance inspection. Conformance inspection shall consist of the examination of 4.4 and tests of 4.5 through 4.5.10.

4.2 Sampling for conformance inspection.

4.2.1 Sampling for examination, measurements and tests performed on entire hose. Unless otherwise specified (see 6.2), for the purpose of sampling, the lot shall consist of all hoses manufactured under the same conditions and at the same time by the same manufacturer. A random sample of hose shall be selected from each lot in accordance with ASQ/ANSI Z1.4 and subject to the examination of 4.4 and tests 4.5.1 through 4.5.6.

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4.2.2 Sampling for the measurements and tests performed on sections cut from hose. The extra length shall be cut from the extra-long hoses (see 3.3) and specimens prepared as required for the tests specified in 4.5.7 through 4.5.9.9.3. A different section may be used for each type of test performed. After cutting the extra lengths from the extra-long hoses, the residual length of the remaining hose shall be 8 feet, 8 inches ±½ inch.

4.2.2.2 Optional method of sampling for tensile properties tests. As an option to performing tensile property tests as specified in 4.5.9.1 through 4.5.9.3 from sections cut from hose (see 4.2.2), the tests may be performed on samples from ASTM laboratory-prepared sheets in accordance with ASTM D412, as approved by the Technical Authority.

4.2.3 Sample for swelling test of rubber friction. A vulcanized sample of the rubber friction used to impregnate the fabric, braid, or spiral wrap reinforcement shall be provided for the friction swelling test (see 4.5.9.4). The sample shall be 6 inches by 6 inches by 0.080 inch ±0.010 inch and shall be the same composition as the friction used in the hose and an equivalent cure.

4.2.4 Sample for flexibility test. One hose assembly shall be selected from each lot, not less than 110 hoses or a combination of smaller lots totaling 110 hoses and subject to the flexibility test of 4.5.10.

4.3 Rejection.

4.3.1 Classification of visual defects and defects in length. Defects shall be classified in accordance with the following:

a. Visual defects shall be classified in accordance with the NAHAD Handbook.

b. Length defects (hoses shorter than specified [see 3.3]) shall be rejected as a major defect. Hoses which are too long shall be cut to the specified length.

4.3.2 Rejection on basis of wall configuration. The lot shall be rejected if the wall configuration is other than that shown on figure 1.

4.3.3 Rejection on the basis of other tests performed on entire hose. The lot shall be rejected if any sample hose (see 4.2.1) fails to meet the requirements for unit weight (see 3.6), proof pressure (see 3.10), or dimensional stability (see 3.10.1).

4.3.4 Rejection on basis of tests performed on hose sections and rubber samples. The lot shall be rejected if the hose sections (see 4.2.2) fail to meet the requirements for burst pressure (see 3.9), tube or cover thickness (see 3.4) or properties of rubber compounds (see 3.11).

4.3.5 Rejection on the basis of flexibility test. The lot shall be rejected if the hose sample fails to meet the requirements for the flexibility test (see 3.8).

4.4 Visual examination. The interior and exterior surfaces of sample hoses (see 4.2.1) shall be examined for the visual defects described in the NAHAD Handbook.

4.5 Tests. Unless otherwise specified herein, tests shall be performed at 80 ±9 °F.

4.5.1 Length. Lengths of sample hoses (see 4.2.1) shall be measured as specified in ASTM D380.

4.5.2 Inside diameter. The i.d. of sample hoses (see 4.2.1) shall be measured as specified in ASTM D380 using plug gages.

4.5.3 Wall configuration and thickness. Wall configuration and thickness of sample hoses (see 4.2.1) shall be measured by vernier caliper and template or suitable dial gage. Measurements shall be made at both ends of the hose.

4.5.4 Unit weight. Unit weights of sample hoses (see 4.2.1) shall be determined by weighing hoses without couplings. Readings shall be accurate to the closest ½-pound.

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4.5.5 Proof pressure. Sample hoses (see 4.2.1) shall be equipped with couplings (see 3.7) and subjected to the proof pressure specified in 3.10. Before starting the test, a 10-pound hydrostatic pressure shall be established in the hose, and the length, o.d., and zero index for the twist shall be measured in accordance with ASTM D380 in preparation for the dimensional stability test specified in 4.5.6. The method shall be in accordance with ASTM D380, except that the hydrostatic pressure rate shall not exceed 1000 lb/in^2 per minute. The proof pressure shall be held for 5 minutes and the hose and fittings examined for leakage or other indication of weakness. At the end of the 5-minute period and while pressure is maintained, the length, o.d., and twist of the hose from zero index shall be measured for the determination of dimensional stability.

4.5.6 Dimensional stability. The effect of the proof test pressure (see 3.10) on the length, o.d. and twist of sample hoses (see 4.2.1) shall be determined in accordance with 3.10.1 and ASTM D380. The gage pressure shall then be reduced to 0 lb/in^2 and the hose left in this condition for 5 minutes. At the end of this period, the diameter and length of the hose shall be measured.

4.5.7 Burst pressure. The hose section obtained in accordance with 4.2.2 shall be equipped with couplings in accordance with 3.7. The burst pressure of this section in the straight condition shall then be determined in accordance with ASTM D380.

4.5.8 Tube and cover thicknesses. Thicknesses of the tube and cover shall be measured on specimens removed from each end of the hose section obtained in accordance with 4.2.2. Thicknesses shall be determined in accordance with the procedure given in ASTM D380. Respective values found for the tube thickness shall not be averaged, but shall be regarded as independent measurements, each of which should meet the requirement. Respective values for the cover thicknesses shall be treated in the same manner.

4.5.9 Properties of rubber components. Properties of rubber components, except the swelling in testing fuel of the friction rubber on the reinforcing fabric, shall be determined on longitudinal specimens prepared from the hose section obtained in accordance with 4.2.2. After separating and buffing smooth, the tube and cover samples shall be permitted to rest not less than 30 minutes before preparing test specimens. Swelling in testing fuel of the friction rubber shall be determined using specimens cut from the sample obtained in accordance with 4.2.3.

4.5.9.1 Initial tensile properties. Initial tensile strength and ultimate elongation shall be determined in accordance with ASTM D412.

4.5.9.2 Tensile properties after immersion. Tensile strength and ultimate elongation shall be determined using specimens cut in accordance with die C of ASTM D412. The specimens shall be immersed in reference fuel B of ASTM D471 at a temperature of 73.4 ±3.6 °F for 46 hours ±15 minutes. Tensile strength shall be based on the swollen cross-section area of the specimen.

4.5.9.3 Tensile properties after oven aging. Tensile strength and ultimate elongation after oven aging shall be determined in accordance with ASTM D573, using specimens cut with die C of ASTM D412, except that the aging period shall be 70 hours ±15 minutes at 212 ±2 °F.

4.5.9.4 Volume change after immersion. The change in volume in percent of the original volume after immersion in reference fuel B of ASTM D471 shall be determined on specimens of the tube, cover, and friction prepared and tested in accordance with ASTM D471. The specimens shall be immersed in fuel at a temperature of 73.4 ±3.6 °F for 46 hours ±15 minutes.

4.5.9.5 Initial adhesion. Adhesion of the tube and cover of the reinforcement, and of the plies of reinforcement to each other, shall be determined in accordance with ASTM D413, machine method, except that the required minimum weight shall be supported by the test specimen for 3 minutes without causing a separation of more than 1 inch.

4.5.9.5.1 Test specimens. Test specimens shall be 2½-inch by 8-inch strips cut longitudinally from the hose. For determining adhesion of the cover or tube to the reinforcement, a ¼-inch wide strip of the cover or tube shall be cut and separated from the underlying reinforcement on each side of a 1-inch wide centered test strip. For determining adhesion between plies of the reinforcement, two plies shall be cut through on each side of the center test strip. The plies to be separated shall be at opposing angles.

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4.5.9.6 Adhesion after immersion. The adhesion of the tube and cover to the reinforcement, and of the plies of reinforcement to each other, shall be determined after the test specimens (see 4.5.9.5.1) have been immersed in reference fuel B of ASTM D471 for 46 hours ±30 minutes at 80 ±9 °F. The test procedure for adhesion shall be in accordance with ASTM D413, machine method, except that the wet specimen shall be tested within 15 minutes after it is removed from the fuel. The adhesion values shall meet the requirements of table I.

4.5.9.7 Nonvolatile extractable material.

4.5.9.7.1 Test specimen. Each specimen shall consist of strips about 1/16-inch square by 2 inches long, totaling about 5 grams in weight, cut from buffed pieces of the tube of the hose.

4.5.9.7.2 Procedure. The specimen shall be weighed and placed in an extraction thimble in the extraction apparatus in accordance with ASTM D471. One hundred milliliters (ml) of type I fluid in accordance with ASTM D471 shall be added to the flask. The specimen shall be extracted for 46 hours ±30 minutes at such a rate that 2½ to 3½ minutes are required to fill and empty the extraction cup. The extract shall be poured into a weighed container. The specimen and flask shall be rinsed with 25 ml of fresh type I fluid and the wash added to the extract. The nonvolatile residue shall be determined from the extract in accordance with ASTM D381, except that the evaporation time shall be 45 minutes. Weight of the residue shall be expressed as percent of the original weight of the specimen. The test shall be performed in duplicate and the results averaged.

4.5.9.8 Phosphate plasticizer test. Phosphate plasticizer shall be determined theoretically by compound formula review or qualitatively.

4.5.9.9 Resistance to ozone.

4.5.9.9.1 Test specimens. Two 5-inch lengths shall be cut from the sample length of hose.

4.5.9.9.2 Apparatus. The apparatus shall consist of a test chamber with an ozone generating source outside the chamber capable of maintaining an ozone-air atmosphere in the chamber at the temperature and concentration desired; a circulating air oven, as described in ASTM D573; and two pairs of metal plates, ⅜ inch by 1½ inches by 8 inches each, with bolt holes spaced centrally ½ inch from each end, and provided with bolts and nuts for clamping the specimens.

4.5.9.9.3 Procedure. The specimen shall be placed between the metal plates and clamped so that the distance across one of the loops thus formed is 2 inches. The bulk of the specimen shall be reduced by cutting away the loop that projects from the side opposite the measure loop. Cut edges of the specimen shall be coated with paraffin to seal them from ozone. The clamped specimens shall be conditioned in an air oven for 16 to 24 hours at 100 ±2 °F. Concentration of ozone in the chamber shall be 1.00 ±0.05 parts of ozone per one million parts of air by volume. Specimens shall be examined at the end of this period for cracking of the hose cover using 7× magnification to determine conformance with the requirements specified in 3.9.

4.5.10 Flexibility test. One end of coupled hose shall be supported from a rigid attachment. The hose shall be bent 180 degrees. A torque shall be applied to the free end of the bent configuration and the hose twisted 90 degrees in each direction. The results shall be in conformance with the requirements specified in 3.8.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

6.1 Intended use. Hoses covered by this specification are intended for use with probe fueling system receivers. The hose is located between the probe receiver outlet and the ship fuel system. This reinforced hose supports the rough handling and excessive bending in this application. Existing stock manufactured to MIL-H-24565, dated 3 March 1978 with Amend 1 dated 19 August 1987, is acceptable for use until depleted. Future requisitions for hoses calling out MIL-DTL-24565 should refer to this version of the specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Length of couple hose required (see 3.8).

c. Sampling lot requirements if other than specified (see 4.2.1).

d. Packaging requirements (see 5.1).

6.3 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are procured by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.4 Definitions.

6.4.1 Technical authority. The Technical Authority has the authority, responsibility, and accountability to establish, monitor, and approve technical standards, tools, and processes in conformance to higher authority policy, requirements, architectures, and standards. NAVSEA 05S may be contacted to determine the appropriate Technical Authority for any given requirement related to equipment and systems of U.S. Navy ships and submarines.

6.5 Subject term (key word) listing.

Fueling-at-sea

Hose, fuel

Hose, reinforced

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Preparing activity:
Navy – SH
(Project 4720-2020-021)

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